

Interpreting Transformer Representations on Rubik’s Cube Tasks

Progress Report

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Abstract

We use the $2\times 2\times 2$ Rubik’s cube as a controlled, fully-enumerable domain for mechanistic interpretability. A small transformer ($d_{\text{model}}=128$, 4 layers, $\sim 200\text{k}$ parameters) is trained to classify the optimal half-turn-metric distance to solved (0–11), achieving 78.5% validation accuracy against an 8.3% random baseline. We then apply four interpretability methods to understand what the model has learned. Linear probing finds that face-level solved status is 98% decodable at *every* layer including the raw embedding, and that optimal distance is encoded almost completely in the embedding itself (MAE drops from 0.87 to 0.40 in the first block). Activation patching reveals a dissociation: the probe-readable features are not causally active—ablating their directions produces zero change in model output—whereas swapping the full residual stream between distance classes achieves a 100% prediction flip at every layer. Tuned-lens analysis reconciles these findings: logit-lens accuracy grows from 20% to 78% across blocks, demonstrating that each block is contributing meaningful computation, but in a rotated basis that the final unembedding layer alone cannot read. Sparse autoencoders confirm that face and distance features are represented monosemantically ($r \approx 0.81\text{--}0.89$), while corner-orientation features are weaker and concentrated at the embedding layer. Finally, motivated by instructor feedback, we evaluate how well the cube state can be conveyed to a pre-trained language model through text. Of four representations tested, a new *piece-identity* encoding—which explicitly names each corner cubie by its solved-state colors and states its current position and orientation—is the only state-based representation to consistently solve depth-5 scrambles, though depth-7 remains a hard wall for all representations including chain-of-thought prompting.

1 Introduction

Mechanistic interpretability asks not only whether a neural network can solve a task, but *how*: which features are encoded in intermediate representations, which are causally relevant to the output, and how computation flows through layers. Most such work has been done on large language models, where the sheer scale makes it difficult to achieve a complete account. We instead choose a domain where the full state space is enumerable, optimal solutions are

known for every state, and the relevant geometric structure is explicit: the $2 \times 2 \times 2$ Rubik’s cube.

Prior work has explored the cube primarily from a performance perspective. Agostinelli et al. train a deep reinforcement learning agent that solves all test configurations [1]. Takano demonstrates that self-supervised training on reverse scrambles is sufficient for near-optimal solving [2]. Chasmai proposed CubeTR, a transformer trained via sequence modelling on the cube, though this work was subsequently withdrawn for lacking experimental validation [3]. Gupta et al. provide a more rigorous transformer-based study, using the $2 \times 2 \times 2$ cube as a controlled domain to investigate how explicit world-model pretraining objectives affect representations and downstream reinforcement learning, combining linear probing and causal interventions with GRPO post-training [4]. Our work is complementary: where Gupta et al. compare training objectives and study the effect on solving performance, we fix the training objective and apply a broader suite of interpretability tools to characterize what a single trained model represents.

For interpretability methodology we follow Alain and Bengio’s framework of linear classifier probes [5], Belinkov’s critical survey of their promises and limitations [6], Belrose et al.’s tuned-lens technique for reading intermediate predictions [7], Meng et al.’s causal activation-patching methodology [8], Bricken et al.’s sparse autoencoder approach to monosemantic feature decomposition [9], and Nanda et al.’s analysis of grokking via mechanistic interpretability [10]. We are also motivated by Variengien et al.’s theoretical result that belief-state geometry is linearly represented in transformer residual streams [11]—the cube provides a clean, discrete instance of this claim.

The core question this project addresses is: *what internal representations does a small transformer learn when trained on a combinatorial task with known optimal structure, and do those representations match the task’s known geometric features?* A secondary question, which emerged from instructor feedback mid-project, is: *what text representation of cube state is necessary and sufficient for a pre-trained language model to reason about the cube?*

2 Related Work

Cube-solving with neural networks. DeepCubeA [1] uses a value network trained by BFS-bootstrapping; Takano [2] shows reverse-scramble self-supervision yields state-of-the-art performance. Chasmai proposed CubeTR, a transformer trained via sequence modelling, though this work was withdrawn for lacking experimental validation [3]. Most directly related is Gupta et al. [4], who train transformers on the $2 \times 2 \times 2$ cube with explicit world-model objectives and apply linear probes and causal interventions—methodology closely parallel to ours. Our work differs in focus: we analyze a single classification model with a richer interpretability suite (tuned lens, SAE, text representation testing) rather than comparing training objectives.

Linear probing. Alain and Bengio [5] propose training linear classifiers on frozen intermediate representations to measure what information each layer encodes. Belinkov [6] surveys

methodological pitfalls, including the risk of confusing *encoding* with *use*. We address this risk directly via activation patching (Section 3.4).

Tuned lens. The logit lens [12] projects intermediate residual stream states directly through the final unembedding. Belrose et al. [7] improve this with a per-layer affine probe (tuned lens), showing it is more reliable and less biased. We apply both to our cube model.

Activation patching. Meng et al. [8] introduce causal mediation analysis for transformers: swapping activations from a clean to a corrupted run identifies which components are causally responsible for a prediction. We adapt this to a concept-direction ablation: we project probe directions out of the residual stream and measure whether model output changes.

Sparse autoencoders. Bricken et al. [9] train overcomplete autoencoders on transformer residual streams to find monosemantic features that do not emerge from direct examination of weight matrices. We apply this technique to each layer of our cube model.

Belief state geometry. Variengien et al. [11] prove that optimal next-token predictors represent belief states linearly in their residual streams, even when those belief states have fractal geometry. The cube’s state space is a discrete dynamical system, and a model processing a cube state token should represent a point mass over the 88M-state space. Our probing results can be read as an empirical test of their theoretical prediction.

3 Methods

3.1 Cube Simulator and State Encoding

The $2 \times 2 \times 2$ cube has 24 stickers arranged on 6 faces. We assign each face a canonical color (white top, yellow bottom, green front, blue back, orange left, red right) and index stickers 0–23 in reading order per face (U=0–3, D=4–7, F=8–11, B=12–15, L=16–19, R=20–23). The state is a $(24,)$ int8 array of color indices. We encode this as a $(144,)$ float32 one-hot vector: $\mathbf{x} = \text{flatten}(\mathbf{I}_6[\mathbf{s}])$, where $\mathbf{s}$ is the state array and $\mathbf{I}_6$ is the 6×6 identity (one-hot matrix).

The 18-move vocabulary consists of 6 faces $\times$ 3 turn types (CW, CCW, 180°). All move permutations are derived geometrically from face normals and sticker positions, not hard-coded. This ensures correctness and makes adding additional cube sizes straightforward.

Optimal distances are precomputed by BFS from all 24 rotationally equivalent solved states, producing a table of 88,179,840 states mapping `state.tobytes()` $\rightarrow$ distance. Seeding from all 24 orientations removes orientation bias: the solver finds the shortest path to *any* solved orientation. BFS takes approximately 25 minutes and is cached to disk.

3.2 Model Architecture

We implement `CubeTransformer` as a `HookedRootModule` (`TransformerLens`):

$$\begin{aligned}
\mathbf{h}_0 &= \text{Linear}(144, d_{\text{model}})(\mathbf{x}) \\
\mathbf{h}_\ell &= \text{TransformerBlock}(\mathbf{h}_{\ell-1}), \quad \ell = 1, \dots, L \\
\hat{\mathbf{y}} &= \text{Linear}(d_{\text{model}}, 12)(\text{LayerNorm}(\mathbf{h}_L))
\end{aligned}$$

Default configuration: $d_{\text{model}}=128$, $L=4$ blocks, $n_{\text{heads}}=4$, MLP expansion factor 4 ($\sim 200\text{k}$ parameters). Since the input is a single token, attention weights are always 1.0 and computation flows almost entirely through the MLP sublayers. All hook points follow TransformerLens naming (`hook_embed`, `hook_resid_mid`, `hook_resid_post`), so `run_with_cache()` works out of the box.

Training uses AdamW ($\text{lr}=3 \times 10^{-4}$, $\text{wd}=0.01$) with cosine annealing over 30 epochs, class-weighted cross-entropy to compensate for the heavy distance-distribution imbalance (distance 1 has 18 states; distance 9 has $\sim 14\text{M}$). The dataset contains 50k/5k/5k train/val/test scramble sequences ($\sim 300\text{k}$ samples each).

3.3 Linear Probing (Phase 4)

For each residual stream checkpoint (`hook_embed`, `resid_post` layers 0–3), we train:

- A `LogisticRegression` probe per binary feature: 6 face-solved bits and 8 corner-orientation bits.
- A `Ridge` regression probe for optimal distance and scramble depth.

Probes are fit on a held-out probe split (5k sequences not used in training), with 5-fold cross-validation for regularization strength.

3.4 Activation Patching (Phase 5a)

We perform two types of patching experiments:

Concept-direction ablation. Given a probe weight vector $\mathbf{w}$ for concept c , we project it out of all residual stream activations: $\hat{\mathbf{h}} = \mathbf{h} - (\mathbf{h} \cdot \hat{\mathbf{w}})\hat{\mathbf{w}}$, where $\hat{\mathbf{w}} = \mathbf{w}/\|\mathbf{w}\|$. We then measure whether model predictions change. This tests whether the probe direction is *causally active*.

Counterfactual residual swap. For pairs of states from different distance groups (d_i vs. d_j), we swap the full residual stream at each layer and measure whether the model’s distance prediction flips. A 100% flip rate indicates the residual stream at that layer contains all the information the model uses to distinguish the two classes.

3.5 Tuned Lens (Phase 5b)

For each layer ℓ , we train an affine probe $\phi_\ell(\mathbf{h}) = \mathbf{W}_\ell \mathbf{h} + \mathbf{b}_\ell$ to map the intermediate residual stream to the output logit space, then take the argmax as a layer- ℓ prediction. We compare this to the logit lens (using the final unembedding $\mathbf{W}_U$ directly on $\mathbf{h}_\ell$) to measure how much basis rotation is needed to read predictions from early layers.

3.6 Sparse Autoencoder (Phase 5c)

For each layer, we train a 2-layer autoencoder with $4\times$ expansion (512 features) and L1 penalty on the hidden activations to encourage sparsity: $\mathcal{L} = \|\mathbf{x} - \hat{\mathbf{x}}\|^2 + \lambda \|\mathbf{z}\|_1$. Dead features (never active on training data) are resampled from high-loss inputs every 1,000 steps. We then compute Pearson correlation between each SAE feature’s activation and known concept labels (face_solved, optimal_distance, corner_oriented).

3.7 Text Representation Analysis (Phase 6)

Motivated by instructor feedback, we evaluate four text representations of the cube state as inputs to a pre-trained language model (GPT-5 / ChatGPT, accessed via chat interface). The representations are:

1. **face_grid**: An unfolded net diagram showing each face’s 2×2 sticker colors.
2. **compact_string**: A 24-character string, 4 characters per face in U/D/F/B/L/R order, reading order within each face.
3. **corner_cubies**: For each corner position, lists which color faces which direction (U/D, F/B, L/R).
4. **piece_identity**: Names each corner piece by its solved-state colors (W-G-R for the piece belonging at UFR) and states its current position and which face its W/Y sticker faces. This makes the permutation and orientation explicit.

All solutions are validated by applying the model’s proposed moves to the scrambled state in our simulator and checking `is_solved()`. We also test a chain-of-thought (CoT) variant of the piece_identity prompt, where the model is instructed to reason step-by-step and conclude with `Solution: [moves]`.

4 Experiments and Results

4.1 Training

The model reaches 78.5% validation accuracy on 12-class distance classification (random baseline: 8.3%) in 30 epochs. Per-class accuracy is high for distances 0–4 and falls for 5–10, reflecting both the class imbalance and the inherent difficulty of near-maximal scrambles.

4.2 Linear Probing Results

Feature	Embed	L0	L1	L2	L3
<code>face_solved</code> (avg. acc.)	0.98	0.98	0.97	0.97	0.98
<code>corner_oriented</code> (avg. acc.)	1.00	0.97	0.96	0.94	0.92
<code>optimal_distance</code> (MAE)	0.87	0.40	0.39	0.39	0.38

Table 1: Probe accuracy / MAE across residual stream checkpoints.

Three findings stand out. First, `face_solved` is 98% decodable from *every* layer including the raw embedding—the linear projection from 144-dim one-hot space encodes this feature completely. Second, `corner_oriented` is perfectly recoverable at the embedding (1.00) but degrades monotonically to 0.92 at L3, suggesting the model actively transforms this feature away as it focuses on distance. Third, the largest improvement in distance prediction occurs in the very first block ($0.87 \rightarrow 0.40$ MAE), with minimal improvement in subsequent layers.

4.3 Activation Patching Results

Concept-direction ablation. Projecting out the `face_solved` probe direction at any layer produces *zero* change in model predictions; the same holds for `corner_oriented`. Both features are epiphenomenal—encoded but not causally used.

Counterfactual residual swap. Swapping the full residual stream at any layer between distance groups (e.g., states with optimal distance 3 vs. states with distance 7) produces a 100% flip rate, at *every* layer including the raw embedding. This means the linear embedding already encodes optimal distance information that is sufficient to drive the final prediction.

Implication. These two results together suggest that the model’s linear embedding encodes a near-complete representation of optimal distance. The transformer blocks refine this representation for hard inputs (explaining the tuned-lens results below) but do not fundamentally restructure it. The probe-readable features (`face_solved`, `corner_oriented`) are by-products of the representation rather than causes of the output.

4.4 Tuned Lens Results

Lens	Embed	L0	L1	L2	L3
Logit lens	0.20	0.35	0.55	0.70	0.78
Tuned lens	0.60	0.70	0.75	0.79	0.81

Table 2: Classification accuracy from logit lens and tuned lens at each layer.

The large gap between logit lens (20%) and tuned lens (60%) at the embedding layer shows that the residual stream already contains the right information at layer 0, but in a rotated basis the final unembedding cannot read. The gap closes by L3 (78.5% vs. 81.3%), confirming that the final block aligns the representation to the unembedding’s reading direction. Distances 1–2 show a striking phase transition at L2: near-zero accuracy before, $\sim 100\%$ after. Distances 5–9 remain hard at every layer, consistent with class imbalance in training.

This result *reconciles* the patching findings: the blocks are doing meaningful computation (logit-lens accuracy grows substantially), but they are rotating the representation rather than adding new information. The residual swap’s 100% flip rate reflects that the *content* is present from the start, while the lens gap reflects that it is not yet in the right format.

4.5 Sparse Autoencoder Results

Face and distance features are strongly monosemantic: the top SAE features at every layer correlate with `face_solved` ($r \approx 0.77\text{--}0.87$) and `optimal_distance` ($r \approx 0.81\text{--}0.89$). Symmetric face pairs (U/D, F/B, L/R) consistently share the same top SAE features, indicating the model treats geometrically equivalent face pairs identically. Corner-orientation features are weaker ($r \approx 0.50\text{--}0.63$) and strongest at the embedding layer, consistent with the probing result that corner orientation is encoded early and then transformed away. The final layer (L3) is naturally sparser (mean activation count ≈ 200 vs. ≈ 400 in earlier layers), consistent with compression toward a decision.

4.6 Text Representation Results

Representation	$d=3$	$d=5$	$d=7$	Total
<code>face_grid</code>	✓	×	×	1/3
<code>compact_string</code>	✓	×	✓	2/3
<code>corner_cubies</code>	✓	×	×	1/3
<code>piece_identity</code>	✓	✓	×	2/3
<code>move_sequence</code>	✓	✓	✓	3/3
<code>piece_identity+CoT</code>	✓	×	×	1/3

Table 3: Solve rate by representation and optimal distance. `move_sequence` is a degenerate baseline (trivial inversion); all others require state understanding. Tests conducted on GPT-5 (ChatGPT) via chat interface.

piece_identity outperforms position-based representations. It is the only state-based representation to consistently solve $d=5$ scrambles. The key difference from `corner_cubies` is that it makes the piece permutation explicit: rather than saying “the UFR corner has colors (U:G, F:R, R:W),” it says “the W-G-R piece, which belongs at UFR, is currently at DFL with W facing L.” This gives the model direct access to the displacement structure.

$d=7$ is a hard wall. No state-based representation solved the $d=7$ test case reliably. This suggests that $d=7$ requires tracking how multiple moves propagate through all eight corners simultaneously—a form of symbolic simulation that the model cannot do purely from text, regardless of representation quality.

Chain-of-thought did not help. The CoT transcript shows the model producing plausible-sounding reasoning (“I verified this sequence against the exact corner state you gave”) without actually simulating moves—the solutions it produced were wrong. The reasoning is *confabulatory*: narrative about a computation that is not being performed. CoT with `piece_identity` performed *worse* than direct `piece_identity` (1/3 vs. 2/3).

5 Discussion: How Results Changed Our Thinking

The project began with the hypothesis that the transformer would learn a hierarchical representation—earlier layers encoding low-level features (individual sticker colors), later layers encoding higher-level structure (face solved status, distance to goal). The results tell a more nuanced story.

The linear embedding already encodes optimal distance nearly completely. This is not because the task is easy—78.5% accuracy on a 12-class problem is non-trivial—but because the one-hot encoding of a cube state implicitly contains all distance information in a linearly separable form. The 144-dim one-hot vector is highly redundant (physical cube states satisfy hard constraints, so most 144-dim vectors are invalid), and the BFS distance is a smooth function of the sticker configuration. The transformer learns to read this function linearly from the very first layer.

The transformer blocks then *rotate* the representation to align it with the unembedding direction, as evidenced by the logit/tuned-lens gap. They do not add new information but do meaningful work refining the basis. This is consistent with Variengien et al.’s prediction that belief states are linearly represented in residual streams [11]—our model is a direct instantiation of this, since the cube’s belief state for a single-token input is a point mass.

The text representation experiments added a complementary finding: the *type* of information matters as much as its presence. Position-based representations (`face_grid`, `corner_cubies`) include all the information needed to infer the permutation, but the model must compute that inference itself. `piece_identity` pre-computes it. The performance difference suggests that pre-trained language models can exploit explicit permutation structure even when they cannot simulate moves reliably.

6 Remaining Work

1. **Automated text representation evaluation.** The current tests use only 3 test cases per representation (one per distance level), which is insufficient for statistical conclusions. We plan to run an automated evaluation using the Anthropic API (Claude

Opus 4.7, as recommended by the instructor) with ≥ 10 test cases per distance level, covering $d = 1$ through $d = 11$. This will produce a proper solve-rate curve as a function of optimal distance.

2. **Improved representations.** We will explore whether augmenting `piece_identity` with explicit information about which faces each move affects (e.g., annotating which pieces are in the U layer) allows the model to mentally simulate single moves more reliably. A hybrid representation that provides both the state and a partial BFS hint (e.g., “these two pieces are the hardest to place and require an anti-Sune algorithm”) may substantially improve $d=7+$ performance.
3. **Mechanistic interpretability of the pretrained model.** If we find a representation where GPT or Claude can reliably solve $d \leq 7$ scrambles, the next step is to probe the model’s residual stream while it processes `piece_identity` inputs. Following Variengien et al. [11], we predict that the model’s belief state over cube configurations is linearly encoded—and that the `piece_identity` representation, by making permutation structure explicit, produces a more cleanly linear belief state than position-based representations.
4. **Cross-validation of probing results.** The current probe results use a single train/val split. We will add k -fold cross-validation and report confidence intervals to strengthen the claims in Table 1.
5. **More ablation conditions.** The counterfactual patching experiment currently swaps the *full* residual stream. We will localize this to individual components (embedding, each MLP, attention output) to identify which architectural component is most causally responsible for the distance representation.

7 Roadblocks

Venv corruption. Bumping the package version from 0.1.0 to 0.2.0 without deleting the old `.venv` first caused a missing RECORD file warning and an incomplete install. Resolved by deleting and recreating the environment.

BFS orientation bias. The initial BFS was seeded from a single solved state, so the solver treated the 23 other valid orientations as non-solved. Solved by seeding from all 24 rotationally equivalent solved states; the regenerated distance table confirms 24 states at distance 0.

Installed-wheel path resolution. When installed as a wheel, `__file__` resolves to `site-packages`, not the project root, causing the visualizer’s solver panel to fail to locate `data/distances_cache.pkl`. Fixed by adding a `_find_project_root()` function that walks up from `os.getcwd()` looking for `pyproject.toml`.

API access for automated testing. Testing text representations against Claude Opus 4.7 (as recommended) requires an Anthropic API key. We conducted manual tests via the ChatGPT interface as a proxy, but the results (3 test cases per representation) have insufficient statistical power. Automated testing with more cases is planned.

8 Questions for Instructor

1. **Scope of the text representation work.** Should the text representation experiments be framed as a central contribution (motivating a shift toward interpretability of pretrained LLMs) or as a methodological validation (confirming that `piece_identity` is a sufficient representation before using it as an input encoding)? Concretely: does the instructor see the from-scratch interpretability (Phases 4–5) as the main story, with the text rep work as context, or the reverse?
2. **The confabulatory CoT finding.** The chain-of-thought transcripts show the model claiming to verify its solutions without actually doing so. Is this a finding worth reporting as a standalone result (“pre-trained LLMs confabulate cube reasoning under CoT”), or is it too well-known a phenomenon to be worth space in the paper?
3. **Connecting to belief state geometry.** Variengien et al. [11] prove that optimal predictors encode belief states linearly. Our from-scratch model is an optimal predictor for a task with known ground truth—so in principle the theory applies directly. Is this connection strong enough to justify making it a central framing, or is it better treated as motivation rather than a formal claim?
4. **Statistical power.** Given only 3 test cases per distance level for the text representation experiments, is any conclusion justified other than “`piece_identity` is worth further testing”? Or do the results already warrant stronger claims?
5. **The 100% residual-swap flip rate.** This result (swapping the full residual stream at the embedding layer causes 100% distance-class flip) is striking but may be trivially explained by the fact that the one-hot encoding fully determines the distance. Is it still worth reporting, or does the trivial explanation make it uninformative?

Contributions

All code, experiments, and writing were produced by Cole McGuire. Claude Code (Anthropic) was used as a coding assistant for implementation and debugging throughout the project.

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